

WEEK 3

LANGUAGE, DATA & AI READINESS

Natural Language Processing • Data Foundations • Capstone
Milestone 1 Prep

TEXTBOOK •
CHAPTER 3

THU • SEP 10, 2026

GRADUATE • 3
HOURS

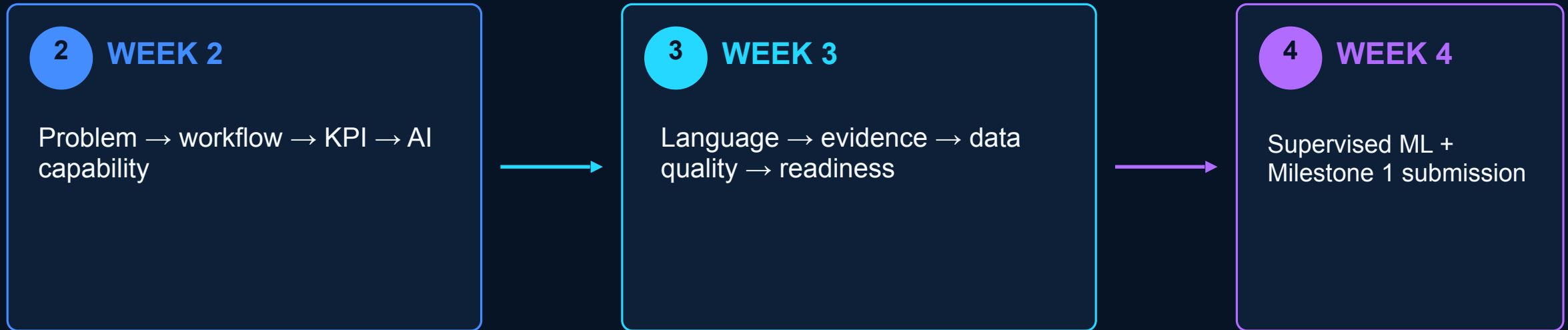
Milestone 1 is due in Week 4 — Thursday, September 17.



WEEK 2 → WEEK 3

From AI capability to usable evidence

Last week: “Which business problem deserves AI?” Today: “What language and data does the system need to work?”



A good AI opportunity is not only valuable. It must also have usable evidence.

TODAY

Week 3 outcomes

By the end of class, students should be able to explain Chapter 3 and use it to strengthen Milestone 1.

1 EXPLAIN

What NLP does and why human language is difficult for machines.

2 DISTINGUISH

Speech recognition from language understanding and downstream NLP tasks.

3 APPLY

Prompting and multimodal Gemini concepts to a business-language task.

4 EVALUATE

Recruiting and other NLP use cases for value, bias, privacy, and failure consequences.

5 MAP

Structured/unstructured data, quality, ownership, labels, and readiness.

6 PREPARE

A complete Milestone 1 Opportunity Proposal for Week 4.

CURRENT AI BRIEFING

AI NOW: three signals to classify

Capability • adoption • governance — keep the headline in context, not in hype.

1 CAPABILITY

Google introduced Gemini 3.8 Flash and 3.8 Flash Cyber for faster agentic workflows and cybersecurity tasks.

Question: What new tasks become practical when speed + reasoning improve?

2 ADOPTION

Large employers are moving AI agents from pilots into everyday employee workflows.

Question: What changes when AI is part of the process—not a separate chatbot?

3 GOVERNANCE

The EU AI Act became broadly applicable on Aug. 2, 2026. Governance is now an operating requirement for many organizations.

Question: Who owns compliance when AI touches a workflow?

PART I

CHAPTER 3 — NATURAL LANGUAGE PROCESSING

Making headway with language: text, speech, multimodality, prompting, and recruitment.

TEXTBOOK • 3.1

Natural Language Processing: AI working with human language

NLP helps systems process, interpret, transform, and generate human language.

READ

Emails • reports •
reviews • contracts

LISTEN

Speech • calls •
meetings • voice
requests

UNDERSTAND

Intent • meaning •
entities • sentiment

TRANSFORM

Summarize •
translate • extract •
classify

GENERATE

Draft • answer •
explain

Business lens: NLP turns language into something a workflow can use.

TEXTBOOK • 3.2

Why language is hard

The same words can mean different things; different words can mean the same thing.

AMBIGUITY

“That charge is sick.”

Meaning depends on context.

CONTEXT

“Cancel it.”

What is “it”? An order?
subscription? meeting?

VARIATION

accents • slang • typos •
jargon • multiple
languages

IMPLICIT MEANING

tone • sarcasm •
urgency • unstated
assumptions

NLP performance depends on context, task definition, and the data/examples the system sees.

TEXTBOOK • 3.2

Text preprocessing & analysis: the textbook mental model

Before a system can use language, it must turn messy input into signals useful for the task.

1 LOWERCASE

Normalize letter case

2 TOKENIZE

Split text into smaller units

3 FILTER

Remove low-information/
filler words

4 STEM

Reduce words toward a
root form

5 POS TAG

Identify grammatical roles

6 ENTITIES

Find names, dates,
amounts, organizations

7 SENTIMENT

Estimate emotional tone

8 CONTEXT / INTENT

Interpret the overall
meaning and goal

Important: this is a useful textbook pipeline—not a claim that every modern NLP/LLM system literally executes these as eight separate modules.

TEXTBOOK • 3.2

A practical NLP task map

Do not start with “NLP.” Start with the specific language task.

CLASSIFY

What category?
intent • topic • spam

EXTRACT

What facts are present?
names • dates • amounts

SENTIMENT

What attitude/tone?
positive • negative • urgency

SUMMARIZE

What is the essence?
meetings • reports • cases

TRANSLATE

Same meaning, new language
customers • global teams

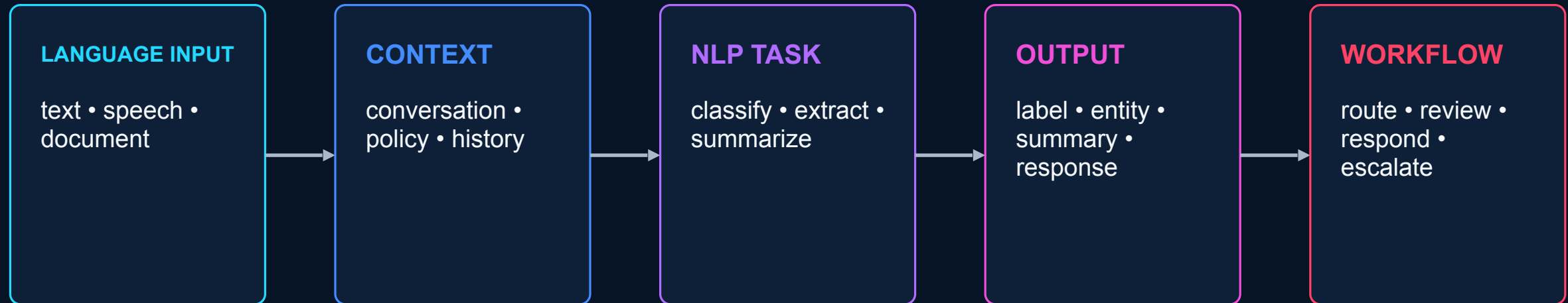
ANSWER / GENERATE

What response is useful?
Q&A • drafts • explanations

TEXTBOOK • 3.2

From language to business action

A useful mental model: input → representation/context → NLP task → workflow decision.



Example: customer voice message → account context → intent + urgency → “billing / high” → priority routing

NLP TASK

Classification: turning language into a category

Useful when the business needs consistent routing, prioritization, or tagging.

INPUT

“I was charged twice and need this fixed today.”

**POSSIBLE OUTPUTS**

Intent: Billing
Issue: Duplicate charge
Urgency: High
Escalation: Yes

Management question: What categories are useful enough to change the workflow?

NLP TASK

Sentiment is useful—but easy to oversimplify

Positive / negative is not the same as business urgency.

MESSAGE A

“I love your product, but my renewal failed and I present to a client in 30 minutes.”

MESSAGE B

“Fine. Cancel it whenever.”

WHY IT MATTERS

Tone, intent, urgency, and risk are different signals.

Do not reduce a workflow to one sentiment score.

A better business design separates the signal you need from the language feature you can estimate.

NLP TASK

Information extraction: finding facts inside language

Turn unstructured text into fields that systems can use.

EMAIL

“Please update PO 8841. The supplier is Northstar Components, the revised total is \$18,740, and delivery moves to Oct. 6.”



EXTRACTED FIELDS

PO: 8841
Supplier: Northstar Components
Amount: \$18,740
Delivery date: Oct. 6

Value comes when extraction reduces manual entry without silently introducing bad data.

NLP TASK

Summarize, translate, and answer

Generative language capabilities can compress or reshape information—but they still require verification.

SUMMARIZE

Meeting transcript → decisions, owners, deadlines

Risk: omitted nuance

TRANSLATE

Customer message → another language

Risk: tone / domain terminology

ANSWER / DRAFT

Question + context → useful response

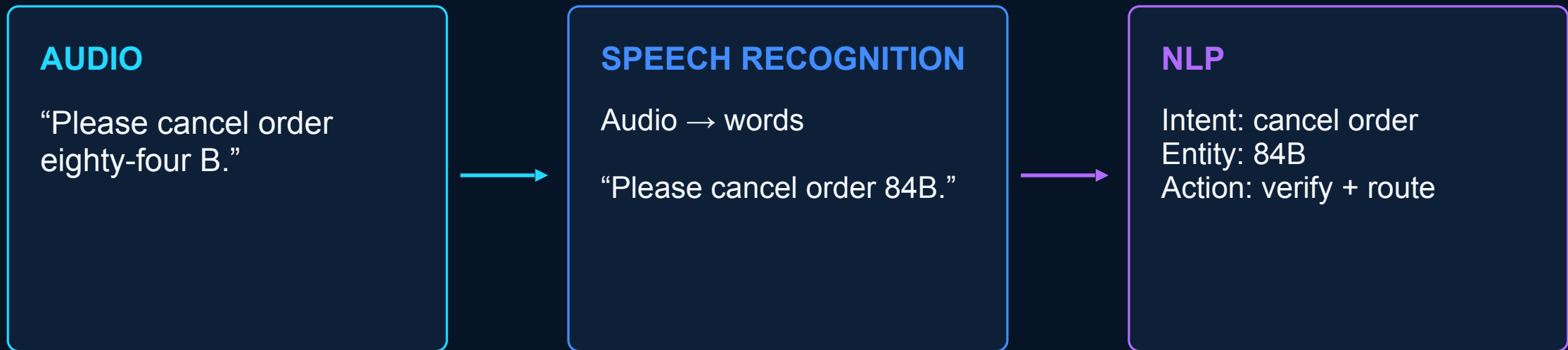
Risk: fabricated or unsupported detail

Fluency is an interface property. Accuracy is an evaluation property.

TEXTBOOK • 3.3

Speech recognition ≠ language understanding

Automatic speech recognition (ASR) converts audio to text; NLP interprets or transforms the language.



Two separate error opportunities: hearing the words incorrectly and interpreting them incorrectly.

TEXTBOOK • 3.3

Gemini as a multimodal example

One model can work across multiple forms of input—text, voice, images, documents, and more.

TEXT

typed request
email
document

VOICE / AUDIO

spoken question
meeting
call

IMAGE / VISUAL

photo
chart
screenshot

COMBINED CONTEXT

“Look at this chart and
explain what I should
tell the client.”

Multimodal does not mean infallible. Each modality adds information—and new failure modes.

TEXTBOOK • 3.4

Prompt engineering does not disappear in voice

Good spoken instructions still define the task, context, expectations, and source.

GOAL

What do you want done?

“Classify this customer request.”

CONTEXT

What situation matters?

“Routing for SaaS customer support.”

EXPECTATIONS

What format / limits?

“Intent, urgency, one-sentence reason.”

SOURCE

What evidence may it use?

“Use only the message I provide.”

The interface changes. The discipline does not.

A reusable business-language prompt

Say it or type it—the structure stays the same.

GOAL

Classify the customer message for routing.

EXPECTATIONS

Return intent, sentiment, urgency, and extracted entities in a table.

CONTEXT

We are a fictional SaaS support team.

SOURCE

Use only the message provided. Do not invent missing facts.

MESSAGE: “I cannot access my account and I present to a client in 30 minutes.”

Evaluation question: Is the output useful for a routing decision—and what could still be wrong?

[SEE IT](#)

LIVE DEMO: one message, two input modes

Use UH Gemini for Education if available. If voice is unavailable, run both examples as text.

1 TYPE

Paste the structured prompt + customer message.

Observe output structure and any assumptions.

2 SPEAK

Use microphone / voice input with the same intent.

Compare transcription and interpretation.

3 CRITIQUE

Did it hear the words correctly?
Did it classify correctly?
Did it invent anything?
Would you automate the route?

Never use real customer, employee, health, financial, or confidential data in the demo.

TEXTBOOK • 3.3

Speech recognition can fail before NLP even begins

Audio quality and language variation can change the downstream business decision.

NOISE

background conversations • poor microphone

ACCENTS / DIALECTS

pronunciation variation • code-switching

JARGON

product names • technical terms • acronyms

NUMBERS / NAMES

84B vs 80-4B • similar names

TURN-TAKING

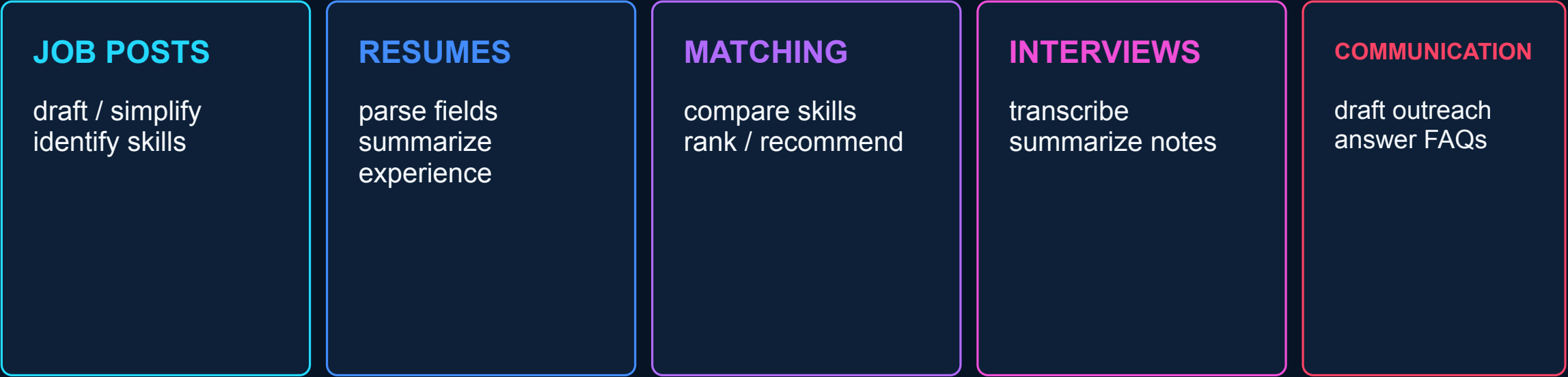
interruptions • overlap • incomplete sentences

PRIVACY

voice can contain sensitive information

Where NLP and GenAI enter recruiting

Recruiting is a useful case because the value is real—and so are the consequences of mistakes.



The higher the decision consequence, the stronger the evidence and human oversight should be.

TEXTBOOK • 3.5

Recruiting: value, evidence, and risk must travel together

A faster hiring workflow is not automatically a better hiring workflow.

POTENTIAL VALUE

- less repetitive screening
- faster candidate communication
- consistent extraction
- recruiter capacity

RISKS

- historical bias
- proxy discrimination
- privacy / sensitive data
- false confidence
- explainability

CONTROL IDEAS

- job-related criteria
- human review
- audit outcomes by group
- data minimization
- appeal / override

Manager question: What evidence would justify using AI at each step of the hiring process?

MINDTAP-STYLE RETRIEVAL

Chapter 3 checkpoint

Explain the idea in your own words—no memorized definitions.

A NLP

What does NLP enable a business system to do with human language?

B SPEECH

What is the difference between speech recognition and language understanding?

C MULTIMODAL

Why can combining voice, text, and visuals help—and what new risks appear?

D PROMPTING

How do Goal, Context, Expectations, and Source improve a voice/text request?

E RECRUITING

Why does faster screening not necessarily mean better hiring?

PART —

10-MINUTE BREAK

When we return: language is data → data quality → AI readiness → Milestone 1.

BE BACK IN 10

PART II

DATA FOUNDATIONS & THE AI FACTORY

Course roadmap: structured/unstructured data, quality, labels, ownership, privacy, representativeness.

COURSE EXTENSION

Language is data—usually unstructured data

AI projects often combine clean database fields with messy documents, messages, audio, and images.

STRUCTURED DATA

Rows / columns / defined fields

customer_id • date • amount • product • outcome

Easy to filter and aggregate.

UNSTRUCTURED DATA

Free-form content

email • PDF • chat • call audio • image • video

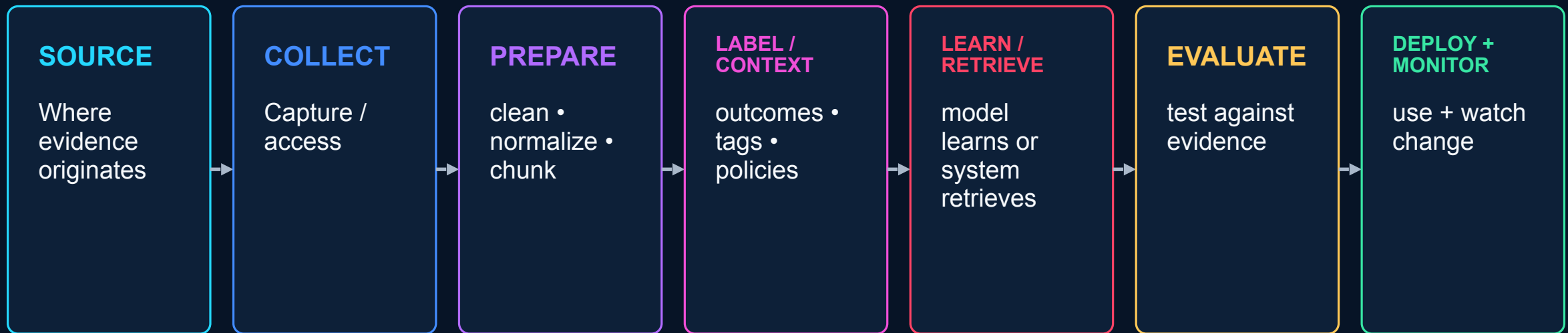
Meaning must be interpreted or extracted.

Most useful business AI combines both.

COURSE EXTENSION

The AI factory: evidence must move through a pipeline

A business AI system depends on repeatable data and evaluation processes.



The pipeline is also where many AI failures begin: inaccessible data, weak labels, stale evidence, or poor monitoring.

COURSE EXTENSION

What kind of evidence does the task need?

Match evidence to the question—not every AI project needs the same data.

PREDICTIVE ML

Historical examples + outcomes

Example: customers + whether they churned

LANGUAGE / NLP

Text or audio + task labels / evaluation examples

Example: support messages + intent

KNOWLEDGE ASSISTANT

Trusted documents + questions + expected grounded answers

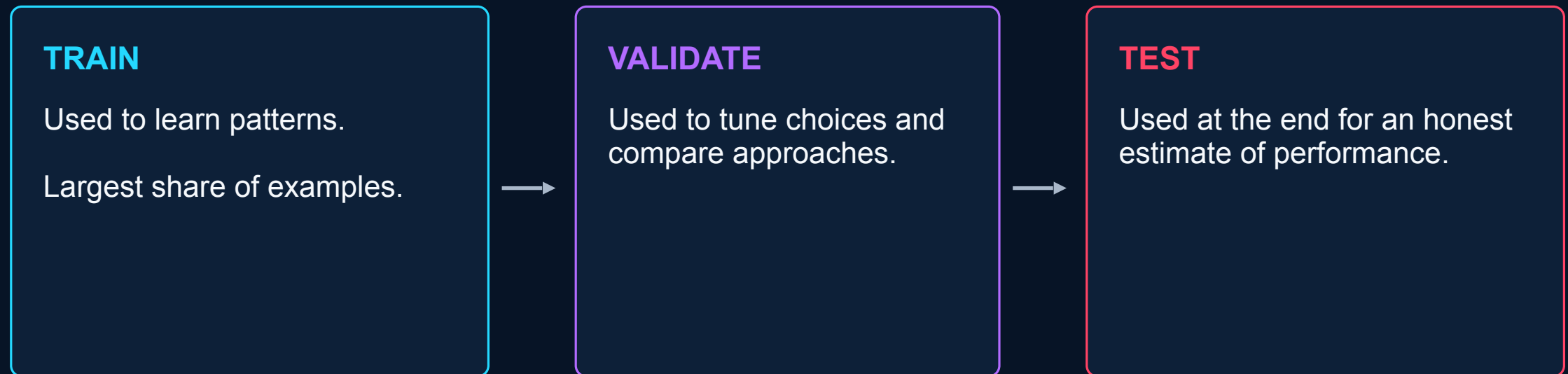
Example: policies + test questions

Data readiness is task-specific. “We have lots of data” is not a feasibility argument.

COURSE EXTENSION

Training, validation, and test data

Keep the final test separate so performance is measured on examples the model did not learn from.



If the test set influences training decisions, the final score can look better than reality.

COURSE EXTENSION

Labels and ground truth: what counts as the “right” answer?

Many AI projects quietly inherit human judgment through their labels.

EXAMPLE 1

Email text → “Billing”

Who decided the category?

EXAMPLE 2

Applicant → “Strong candidate”

What criteria produced that label?

EXAMPLE 3

Transaction → “Fraud”

Was it confirmed—or only suspected?

Bad labels teach the wrong target consistently.

COURSE EXTENSION

Six dimensions of data quality

A dataset can be large and still be unfit for the task.

ACCURATE

Correct?

COMPLETE

Missing key fields/examples?

CONSISTENT

Same definitions/formats?

RELEVANT

Matches the decision?

TIMELY

Current enough?

TRACEABLE

Source/provenance known?

Add one more question: representative of the people and conditions where the system will be used?

COURSE EXTENSION

Representativeness: who and what is missing?

A model can perform well on the dataset and poorly in the population that actually matters.

CUSTOMERS

Training data overrepresents long-term customers and underrepresents new users.

LANGUAGE

Examples reflect one accent, language variety, writing style, or region.

TIME / CONDITIONS

Historical behavior came from a different policy, market, or economic environment.

Ask: “Representative of what deployment population, under what conditions?”

COURSE EXTENSION

Ownership, access, privacy, and permissions

“The data exists” is different from “we can legally and responsibly use it.”

OWNER

Who controls the source?

ACCESS

Can the project team obtain it?

PRIVACY

Does it contain sensitive or personal information?

PERMISSION

Is this use allowed under policy, consent, contract, and tool rules?

For class projects: use public, fictional, synthetic, or instructor-approved data. Do not upload sensitive real data.

COURSE EXTENSION

Data readiness scorecard

Green does not mean “perfect.” It means feasible enough to investigate responsibly.

SOURCE

We know where evidence comes from.

ACCESS

We can obtain/use it appropriately.

QUALITY

It is accurate/relevant enough.

COVERAGE

It represents key cases/users.

TARGET

We know the label/outcome or evaluation criterion.

PRIVACY

Risks and permissions are understood.

If 2+ areas are red, Milestone 1 should usually be reframed before prototype work.

PART III

CAPSTONE MILESTONE 1 — OPPORTUNITY PROPOSAL

Due in Week 4 • Thursday, September 17 • approval gate before prototype work.

TEAMS ARE FORMED. THIS WEEK = REFINE + VALIDATE.

STONE 1 • 4% OF COURSE

Exactly what Milestone 1 must contain

1–2 page Opportunity Proposal + one-slide executive summary. No prototype required yet.

1 PROBLEM + STAKEHOLDER

Specific pain; who experiences it?

2 CURRENT WORKFLOW

What happens today? Where is the friction?

3 BASELINE + KPI

Current performance + measurable target.

4 AI FIT

Primary capability and why it may help.

5 EVIDENCE / DATA PATH

Likely sources, access, quality, and constraints.

6 NON-AI ALTERNATIVE

Simpler solution considered.

7 RISK + CONTROL

One meaningful failure + initial guardrail.

8 RECOMMENDATION

Investigate • reframe • reject.

Approval standard: meaningful + measurable + evidence-feasible + appropriately scoped.

MILESTONE 1

Weak idea vs. approvable opportunity

The difference is not how impressive the technology sounds—it is how clear the business case is.

WEAK

“We want to build an AI chatbot for HR using Gemini.”

Missing: specific user/problem • baseline • KPI • evidence path • risk boundary • non-AI alternative

STRONGER

“Employees spend ~15 minutes locating benefits answers, while HR handles ~600 repetitive questions/month. Investigate whether a grounded knowledge assistant can cut search time below 5 minutes without increasing policy-error rate.”

Notice: the stronger statement does not depend on a tool name.

MILESTONE 1

Example Milestone 1 executive summary

A good summary lets an executive understand the opportunity in 60 seconds.

PROBLEM

Employees lose time searching benefits policies; HR repeats common answers.

BASELINE / KPI

Baseline: ~15 min search; ~600 repeated HR questions/month.
Target: <5 min; policy error not worse.

AI FIT

Language Q&A / knowledge assistance. AI may help interpret questions and retrieve relevant policy.

EVIDENCE

Policy PDFs; anonymized question categories; timing/volume estimates.
Use fictional/public data for prototype.

ALTERNATIVE

Improve policy search/navigation + FAQ taxonomy without GenAI.

RISK / CONTROL

Risk: unsupported policy answer.
Control: citation + “I don’t know” + HR escalation for ambiguous cases.

Recommendation: INVESTIGATE — strong recurring problem with measurable value and controllable scope.

BEFORE WEEK 4

At-home activity

Individual • ~30–45 minutes • Use UH Gemini for Education with only the fictional data provided.

A MANUAL

Classify six fictional customer messages: intent, sentiment, urgency, entities.
(Generate them manually, not the AI-generated, we are going to evaluate AI based on our prompt)

B GEMINI

Run the provided structured prompt.
Compare AI output with your manual judgment.

C EVALUATE

Identify two disagreements/errors and explain business consequence.

Optional voice extension: dictate one fictional message and compare the transcript with the original.

LEAVE WEEK 3 WITH FIVE IDEAS

Executive takeaway

Chapter 3 + data readiness + capstone preparation in one picture.

1 LANGUAGE IS DATA

NLP turns messy language into labels, facts, summaries, or responses.

2 SPEECH HAS TWO STEPS

Hearing words correctly and interpreting them correctly are separate risks.

3 PROMPTS DEFINE THE TASK

Goal + Context + Expectations + Source improve clarity—but output still needs evaluation.

4 DATA MUST BE FIT FOR PURPOSE

Quality, labels, representativeness, access, and privacy shape feasibility.

5 MILESTONE 1 = APPROVAL GATE

Do not prove the AI works yet. Prove the opportunity is worth investigating.

BEFORE WEEK 4 → MindTap Chapter 3 • At-home NLP activity • Milestone 1 due Thursday, Sept. 17